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Machine learning and SHAP values for predicting coronary artery disease risk in Xinjiang, China

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Abstract

Background Accurate individual risk assessment is crucial for guiding and improving the prevention of atherosclerotic cardiovascular disease (ASCVD). Existing prediction models are primarily derived from Western Caucasian and Chinese Han populations. Our objective is to develop and validate an interpretable machine learning (ML) model based on biomarkers for predicting coronary artery disease (CAD) risk among multi-ethnic patients in Xinjiang, China.

Methods This retrospective cohort study enrolled patients who underwent coronary angiography or coronary computed tomography angiography (CCTA) at the First Affiliated Hospital of Xinjiang Medical University. The cohort was divided into training, validation, and test sets. Feature selection was performed using logistic regression and LASSO, followed by prediction model development with six machine learning algorithms (XGBoost, RF, MLP, SVM, KNN, AdaBoost). Predictive performance was evaluated using the area under the receiver operating characteristic curve (AUROC) as the primary metric to identify the optimal algorithm. The selected algorithm was further validated on both the validation and testing sets. Shapley Additive Explanations (SHAP) were applied to quantify each feature's contribution to CAD risk prediction, generating individualized risk explanations. Furthermore, the model's calibration was assessed using calibration curves and the Brier score. Its clinical utility was evaluated through decision curve analysis and was benchmarked against the established SCORE2 Asia Pacific risk model.

Results This study enrolled 7655 male and 4461 female participants, divided into training, validation, and test sets in a 7:1.5:1.5 ratio. XGBoost demonstrated optimal performance in both cohorts: the male model achieved AUROCs of 0.845 (95% CI: 0.834–0.855), 0.814 (0.789–0.839), and 0.826 (0.802–0.850) in the training, validation, and test sets, respectively, while the female model attained values of 0.817 (0.802–0.832), 0.759 (0.721–0.796), and 0.786 (0.751–0.821). Male CAD risk was significantly associated with advanced age, multiple abnormal clinical indicators (elevated creatinine, total cholesterol, lipoprotein(a), etc., and decreased HDL-C), hypertension, and diabetes, with higher risk observed in Kazakh and Hui ethnicities, whereas higher education and married status served as protective factors. In females, hypertension was the strongest predictor, while elevated uric acid, systolic blood pressure, fasting blood glucose, along with histories of hypertension and diabetes increased risk; married status and higher education similarly exhibited protective effects. The prediction model demonstrated favorable clinical utility and accuracy

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in both cohorts, with calibration significantly enhancing predictive performance. Compared to the SCORE2 Asia Pacific risk model, our model exhibited superior discriminatory ability (male: 0.826 vs. 0.662; female: 0.786 vs. 0.720) and improved calibration.

Conclusions Machine learning models can provide personalized and highly accurate predictions of CAD risk. The interpretability of these models facilitates the identification of modifiable risk factors in individual patients, offering valuable insights to enhance primary prevention and management of cardiovascular disease in the Xinjiang region.

Keywords Machine learning model, Coronary artery disease, SHAP, Lp(a), Xinjiang, SCORE2 Asia Pacific risk model

Introduction

The World Health Organization (WHO) has identified cardiovascular disease (CVD), particularly coronary artery disease (CAD) as the leading cause of premature mortality worldwide, accounting for 17.9 million deaths annually—approximately 31% of global mortality [1, 3]. Projections indicate that by 2030, CAD alone will contribute to over 23 million deaths, representing nearly 30.5% of the world's population [4, 5]. The burden of CVD is especially severe in China. According to the China Health Statistics Yearbook 2021, CVD exhibits the highest incidence and mortality rates among urban and rural populations, surpassing even cancer [6]. While invasive coronary angiography remains the gold standard for CAD diagnosis [4], its widespread adoption is hindered by high equipment costs, stringent technical requirements for practitioners, and associated complications (e.g., radiation exposure) [7]. These limitations render angiography largely inaccessible in primary healthcare settings and economically disadvantaged regions [6].Early diagnosis and timely intervention for CAD patients are critical to preventing sudden cardiac death and reducing the financial burden of advanced treatments. Consequently, there remains an urgent, unmet clinical need for rapid, non-invasive, and cost-effective CAD screening tools [4, 7].

Since the development of the first CHD risk equation in 1976 by the Heart Disease Prevention Study, numerous CVD risk assessment tools have been established to guide public health and clinical practice across diverse populations. These include: the Systematic Coronary Risk Evaluation (SCORE) model for European populations, the QRISK algorithm in the UK, and the Pooled Cohort Equations (PCE) for ASCVD endorsed by the American College of Cardiology/American Heart Association guidelines [8–10]. However, these risk prediction models were all derived from Western population cohorts, significantly limiting their generalizability to other ethnic groups. While China has developed its own risk prediction models (CHINA-PAR 10-year and lifetime risk models) to address national characteristics [11], substantial challenges remain due to vast geographical disparities across China's regions, its multi-ethnic diversity with varying genetic backgrounds [12, 13], and heterogeneous life-style and dietary patterns. These variations render generic models unsuitable for direct application across different Chinese populations [14], particularly in ethnically diverse regions like Xinjiang.

Machine learning (ML) has exhibited superior predictive capability over conventional logistic regression approaches [5, 15, 16], primarily due to its ability to capture complex nonlinear associations between risk factors and clinical outcomes, operate without strict data distribution assumptions, and adapt flexibly to diverse healthcare data types [17]. Building on these strengths, this study aimed to develop and validate an interpretable, biomarker-based ML model for predicting cardiovascular disease (CAD) risk among multi-ethnic populations in Xinjiang, with the model designed to enhance diagnostic accuracy and risk stratification in primary care settings, facilitate timely individualized clinical interventions, reduce coronary artery disease (CAD) mortality and improve patient outcomes, and minimize unnecessary medical procedures through precise risk assessment.

Materials and methods

Study participants

This is a retrospective study that selected Han, Uyghur, Kazakh, and Hui patients who underwent coronary angiography while hospitalized in the Department of Cardiology at the First Affiliated Hospital of Xinjiang Medical University between December 2022 and April 2025 as the study subjects. Eligible participants met guideline-defined cardiovascular disease (CVD) diagnostic criteria, were aged ≥ 18 years of age, and provided written informed consent. We excluded patients with severe comorbidities (including cerebrovascular, hepatic/renal, infectious, malignant, or hematologic diseases), mental illness, pregnancy/lactation, contraindications to angiography, or incomplete clinical data. The study protocol was approved by the hospital's Institutional Review Board (Approval Nos.: 2022TSYCLJ0030; 2024E02043) in accordance with the Declaration of Helsinki, with all participants providing informed consent.

Data collection

We systematically collected comprehensive baseline characteristics from clinical records, including: (1) demographic data (age, sex); (2) anthropometric measurements (height, weight, BMI, admission blood pressure, and heart rate); (3) medical history (hypertension, diabetes, smoking, and alcohol use); (4) laboratory test results; and (5) coronary imaging findings (computed tomography angiography [CTA] and angiography reports). Laboratory assessments comprised fasting blood glucose (FBG), lipid profiles (total cholesterol [TC], triglycerides [TG], low-density lipoprotein cholesterol [LDL-C], high-density lipoprotein cholesterol [HDL-C], and lipoprotein(a) [Lp(a)]), uric acid (UA), creatinine (Crea), and urea nitrogen (Urea). Following standardized protocols, venous blood samples were collected after an overnight fast within 24 h of admission. All specimens were processed by the hospital's central laboratory using standardized centrifugation procedures and analyzed on Roche COBAS 8000 automated analyzers with manufacturer-provided reagents (Roche Diagnostics, Switzerland) to ensure measurement consistency. All predictor variables consisted of clinical data collected at patient presentation, prior to the clinician's final diagnosis.

Definitions and evaluation criteria of indicators

In this study, coronary artery disease (CAD) was diagnosed when coronary angiography revealed $\geq 50\%$ stenosis in either the left main coronary artery or at least one major epi-cardial branch. Educational attainment was classified into three levels: primary education (elementary school or below), secondary education (middle school or vocational secondary education), and higher education (college or above). Diabetes mellitus was defined as either self-reported physician-diagnosed diabetes or current use of glucose-lowering drugs or insulin, while hypertension was defined as self-reported physician diagnosis or current antihypertensive medication use. Baseline demographic and clinical characteristics were systematically collected from medical records, including anthropometric measurements, medical history, and comprehensive laboratory test results (fasting blood glucose, lipid profiles, renal function tests, etc.).

Screening of feature variables

We initially evaluated 21 CAD-related candidate risk factors through univariate logistic regression analysis, retaining variables with $p < 0.05$ for further selection. These variables subsequently underwent Least Absolute Shrinkage and Selection Operator (LASSO) regression to identify the most predictive risk factors for CAD development. Variables with non-zero coefficients from the LASSO analysis were then entered into a multivariate logistic regression model for final selection, where only factors maintaining statistical significance ($p < 0.05$) were incorporated into the ultimate predictive model.

Machine learning model construction, performance evaluation, and selection

The dataset was randomly partitioned into training (70%), validation (15%), and test (15%) sets, designated for model training, parameter tuning, and final evaluation, respectively. Using the selected features, predictive models were constructed with six machine learning algorithms: XGBoost, Random Forest (RF), AdaBoost, Multilayer Perceptron (MLP), k-Nearest Neighbors (KNN), and Support Vector Machine (SVM).

To ensure robust model comparison, each candidate model was evaluated using a 10-repeat bootstrap resampling procedure on the training set, with performance metrics calculated as the average across all repeats. Model selection was primarily based on the area under the receiver operating characteristic curve (AUROC) values observed in both the training and validation sets, while comprehensively evaluating four key metrics: AUROC (the primary criterion), accuracy, sensitivity, and specificity. Since the AUROC integrates both sensitivity and specificity and provides an intuitive visualization of overall predictive performance, XGBoost was identified as the best-performing classifier for subsequent in-depth optimization.

Model optimization and SHAP value interpretation

For the optimal XGBoost model identified in the previous section, we implemented systematic hyper-parameter optimization. A grid search was conducted across 648 parameter combinations, with particular focus on seven key parameters: learning rate (0.001, 0.005, 0.01), maximum tree depth (2, 3), subsample ratio (0.6, 0.7), feature sampling ratio (0.6, 0.7), minimum loss reduction (3, 5, 7), L2 regularization (5, 10, 15), and L1 regularization (0.5, 1, 2).The "generalization score" served as our model selection criterion: Validation AUROC—$0.5 \times$(Training AUROC—Validation AUROC). This metric effectively balances model performance against overfitting risk. We employed an early stopping strategy (maximum of 2000 boosting rounds, with early stopping after 100 rounds without improvement) during model training, while continuously monitoring the training process. Upon completion of optimization, we applied SHAP (SHapley Additive exPlanations) analysis to enhance model interpretability. This approach quantifies feature contributions through Shapley value calculations, enabling both global interpretation of feature importance and individualized explanations of how specific factors influence predictions for

particular cases. The final model was comprehensively evaluated using multiple metrics, with the bootstrap method (1000 replicates) employed to calculate 95% confidence intervals, thereby ensuring statistical reliability of the results.

Model calibration and performance validation

Model calibration was performed using Platt scaling on the training set with a gentle strength parameter (0.3) to balance calibration effectiveness and model discrimination. Calibration performance was assessed using the Brier score and calibration curves (Hosmer–Lemeshow test was excluded due to its limitations in machine learning scenarios). The SCORE2 Asia Pacific risk model was also evaluated on our dataset for comparison. Clinical utility was evaluated through decision curve analysis (DCA) and clinical impact curves, while a comprehensive threshold analysis across 99 cut-points (0.01–0.99) was conducted to determine the optimal decision threshold by synthesizing sensitivity, specificity, F1 score, and Youden's index.

Statistical analysis

Statistical analysis and data visualization were conducted using SPSS 27.0 (IBM, Armonk, NY, USA), R 4.4.2 (R Foundation for Statistical Computing, Vienna, Austria), and GraphPad Prism 9.5.0 (GraphPad Software, San Diego, CA, USA). For measurement data adhering to a normal distribution, values were presented as mean ± standard deviation (x ± s), with intergroup comparisons performed via Student's t tests. Skewed-distributed measurement data were summarized as median (first quartile, third quartile) [M (Q1, Q3)], and Mann–Whitney U tests were employed for between-group comparisons. Categorical variables were reported as counts and percentages (%) and analyzed using Pearson's chi-square test, with statistical significance defined as $P < 0.05$.

Univariate logistic regression analysis was first applied to screen candidate feature variables, with variables exhibiting $P < 0.05$ advanced to subsequent LASSO (Least Absolute Shrinkage and Selection Operator) regression for identifying risk factors associated with CAD development. The "glmnet" package in R was utilized to implement LASSO regression, which minimizes prediction error by imposing L1 regularization constraints, forcing some regression coefficients to zero. This technique employs the regularization parameter λ (lambda) for variable selection and model complexity adjustment, making it well-suited for high-dimensional data analysis and risk factor extraction.

Subsequently, multivariate logistic regression analysis was performed on the non-zero coefficient variables identified by LASSO regression, with variables maintaining $P < 0.05$ incorporated into the final predictive model. For machine learning modeling, R packages including "caret", "dplyr", "tidyr", "xgboost", "randomForest", "nnet", "adabag", "e1071", and "class" were employed to conduct model training, cross-validation, hyper-parameter tuning, and confusion matrix computation. Model performance was evaluated using 0.5 as the default classification threshold. Model interpretability was achieved through the "SHAPforxgboost" and "iml" packages, which quantified the contribution of feature variables to prediction outcomes. Visualization results, including receiver operating characteristic (ROC) curves and confusion matrix heatmaps, were generated using the "ggplot2" and "pROC" packages.

Model calibration was performed using Platt scaling (strength parameter $\lambda = 0.3$), with calibration curves plotted using the rms and ggplot2 packages. Clinical utility was assessed through decision curve analysis implemented with the dcurves package, and clinical impact curves were generated using the rmda package.

SPSS 27.0 was primarily used for initial data cleaning, descriptive statistics, and basic hypothesis testing, while GraphPad Prism 9.5.0 was exclusively employed for generating publication-quality statistical graphs. The R 4.4.2 environment was utilized for all advanced statistical modeling, including machine learning, calibration, and clinical utility analysis. All core analyses conducted in SPSS were cross-verified through R to ensure result consistency and reproducibility.

Results

Basic characteristics of participants

A total of 7655 male subjects and 4461 female subjects were included in this study. Of these, 4451 males had coronary artery disease (CAD) and 3104 did not. The largest age group among CAD subjects was 50–59 years (38%). Among male subjects, there were 3276 Han (43%), 2828 Uyghurs (37%), 805 Kazakhs (11%), and 746 Hui (10%) (Table 1). In the female cohort, 1787 had CAD and 2674 did not, with CAD subjects predominantly in the 60–69 years age group (39%). Among female subjects, Han accounted for 45%, Uyghurs 34%, and Kazakhs and Hui each 11% (Table 2). The proportions of hypertension (46% in males, 55% in females) and diabetes (23% in males, 23% in females) in the CAD group were higher than those in the non-CAD group (18% hypertension and 8% diabetes in males; 18% hypertension and 7% diabetes in females), with all $P < 0.05$ (Tables 1, 2). Current smokers were predominantly male, and there was a statistically significant difference between the CAD and non-CAD groups ($P = 0.013$) (Table 1).

Table 1 Baseline data of male subjects

Variables	Total ($n=7655$)	CAD ($n=4551$)	Non-CAD ($n=3104$)	*p* value
Age, *n* (%)				< 0.001
18–29	123 (2)	10 (0)	113 (4)	
30–39	635 (8)	183 (4)	452 (15)	
40–49	1898 (25)	920 (20)	978 (32)	
50–59	2612 (34)	1728 (38)	884 (28)	
60–69	1699 (22)	1201 (26)	498 (16)	
≥ 70	688 (9)	509 (11)	179 (6)	
Ethnic, *n* (%)				< 0.001
Han	3276 (43)	1662 (37)	1614 (52)	
Uyghur	2828 (37)	1798 (40)	1030 (33)	
Kazakh	805 (11)	565 (12)	240 (8)	
Hui	746 (10)	526 (12)	220 (7)	
Marital status, *n* (%)				< 0.001
Married	6449 (84)	3598 (79)	2851 (92)	
Never married	1078 (14)	860 (19)	218 (7)	
Divorced or Widowed	128 (2)	93 (2)	35 (1)	
Education, *n* (%)				< 0.001
Elementary	988 (13)	707 (16)	281 (9)	
Middle	3545 (46)	2300 (51)	1245 (40)	
Higher	3122 (41)	1544 (34)	1578 (51)	
Current smoking, *n* (%)				0.013
Yes	2839 (37)	1636 (36)	1203 (39)	
No	4816 (63)	2915 (64)	1901 (61)	
Current drinking, *n* (%)				0.966
Yes	2398 (31)	1427 (31)	971 (31)	
No	5257 (69)	3124 (69)	2133 (69)	
Hypertension, *n* (%)				< 0.001
Yes	2619 (34)	2073 (46)	546 (18)	
No	5036 (66)	2478 (54)	2558 (82)	
Diabetes, *n* (%)				< 0.001
Yes	1314 (17)	1063 (23)	251 (8)	
No	6341 (83)	3488 (77)	2853 (92)	
BMI, *n* (%),kg/m2				0.002
< 18.5	35 (0)	20 (0)	15 (0)	
18.5–23.9	1554 (20)	888 (20)	666 (21)	
24.0–27.9	3427 (45)	1996 (44)	1431 (46)	
≥ 28	2639 (34)	1647 (36)	992 (32)	
Heart rate, Median (Q1, Q3),beats/min	78 (71, 85)	78 (71, 85)	78 (72, 84)	0.499
SBP, Median (Q1, Q3), mmHg	125 (115, 138)	125 (115, 139)	123 (114, 136)	< 0.001
DBP, Median (Q1, Q3), mmHg	78 (70, 85)	77 (70, 84)	78 (70, 85)	0.002
Crea, Median (Q1, Q3), μmol/L	74.2 (64, 85)	76.9 (67, 87.82)	70 (60.2, 81.1)	< 0.001
Urea, Median (Q1, Q3), mmol/L	5.4 (4.5, 6.4)	5.5 (4.6, 6.6)	5.12 (4.3, 6.1)	< 0.001
UA, Median (Q1, Q3), μmol/L	331 (276, 389.9)	337.3 (287, 394.1)	319.65 (263, 383)	< 0.001
TC, Median (Q1, Q3), mmol/L	4.02 (3.3, 4.8)	3.79 (3.09, 4.69)	4.26 (3.65, 4.93)	< 0.001
TG, Median (Q1, Q3), mmol/L	1.45 (1.01, 2.09)	1.47 (1.03, 2.11)	1.41 (0.98, 2.06)	0.002
HDL-C, Median (Q1, Q3), mmol/L	0.99 (0.84, 1.19)	0.94 (0.81, 1.11)	1.09 (0.92, 1.31)	< 0.001
LDL-C, Median (Q1, Q3), mmol/L	2.64 (2.03, 3.28)	2.49 (1.89, 3.19)	2.81 (2.31, 3.38)	< 0.001
LP(a), Median (Q1, Q3), mg/L	123.25 (57.23, 257.90)	141.13 (66.35, 291.22)	103.15 (49.28, 207.65)	< 0.001
FBG, Median (Q1, Q3), mmol/L	5.1 (4.51, 6.01)	5.22 (4.52, 6.42)	4.99 (4.51, 5.6)	< 0.001

Table 2 Baseline data of female subjects

Variables	Total ($n=4461$)	CAD ($n=1787$)	Non-CAD ($n=2674$)	*p* value
Age, *n* (%)				< 0.001
18–29	40 (1)	2 (0)	38 (1)	
30–39	239 (5)	23 (1)	216 (8)	
40–49	785 (18)	137 (8)	648 (24)	
50–59	1621 (36)	601 (34)	1020 (38)	
60–69	1270 (28)	695 (39)	575 (22)	
≥ 70	506 (11)	329 (18)	177 (7)	
Ethnic, *n* (%)				< 0.001
Han	1989 (45)	731 (41)	1258 (47)	
Uyghur	1509 (34)	623 (35)	886 (33)	
Kazakh	478 (11)	198 (11)	280 (10)	
Hui	485 (11)	235 (13)	250 (9)	
Marital status, *n* (%)				< 0.001
Married	3786 (85)	1391 (78)	2395 (90)	
Never married	363 (8)	209 (12)	154 (6)	
Divorced or Widowed	312 (7)	187 (10)	125 (5)	
Education, *n* (%)				< 0.001
Elementary	1071 (24)	579 (32)	492 (18)	
Middle	2085 (47)	893 (50)	1192 (45)	
Higher	1305 (29)	315 (18)	990 (37)	
Current smoking, *n* (%)				0.4
Yes	49 (1)	23 (1)	26 (1)	
No	4412 (99)	1764 (99)	2648 (99)	
Current drinking, *n* (%)				0.948
Yes	68 (2)	28 (2)	40 (1)	
No	4393 (98)	1759 (98)	2634 (99)	
Hypertension, *n* (%)				< 0.001
Yes	1464 (33)	981 (55)	483 (18)	
No	2997 (67)	806 (45)	2191 (82)	
Diabetes, *n* (%)				< 0.001
Yes	587 (13)	405 (23)	182 (7)	
No	3874 (87)	1382 (77)	2492 (93)	
BMI, *n* (%),kg/m2				< 0.001
< 18.5	59 (1)	18 (1)	41 (2)	
18.5–23.9	1504 (34)	509 (28)	995 (37)	
24.0–27.9	1722 (39)	720 (40)	1002 (37)	
≥ 28	1176 (26)	540 (30)	636 (24)	
Heart rate, Median (Q1, Q3), beats/min	77 (71, 84)	77 (70, 83)	78 (72, 84)	0.008
SBP, Median (Q1, Q3), mmHg	125 (115, 139)	130 (119, 144)	122 (112, 134)	< 0.001
DBP, Median (Q1, Q3), mmHg	76 (69, 82)	76 (68, 82)	76 (70, 82)	0.236
Crea, Median (Q1, Q3), μmol/L	62.3 (53, 72.5)	60.51 (52, 70.5)	63.39 (54, 74)	< 0.001
Urea, Median (Q1, Q3), mmol/L	5 (4.2, 5.9)	5 (4.3, 6.1)	4.9 (4.1, 5.9)	< 0.001
UA, Median (Q1, Q3), μmol/L	277.7 (231.6, 331)	269 (228.7, 318.1)	283.55 (233.2, 341.95)	< 0.001
TC, Median (Q1, Q3), mmol/L	4.3 (3.61, 5.02)	4.25 (3.49, 5.1)	4.33 (3.67, 4.98)	0.076
TG, Median (Q1, Q3), mmol/L	1.35 (0.98, 1.91)	1.39 (1.03, 1.97)	1.32 (0.94, 1.88)	< 0.001
HDL-C, Median (Q1, Q3), mmol/L	1.14 (0.97, 1.36)	1.12 (0.96, 1.32)	1.15 (0.97, 1.39)	< 0.001
LDL-C, Median (Q1, Q3), mmol/L	2.8 (2.24, 3.41)	2.74 (2.14, 3.45)	2.84 (2.31, 3.38)	0.023
LP(a), Median (Q1, Q3), mg/L	127.3 (61.56, 255.32)	150.94 (73.3, 311.4)	112.7 (56, 222.86)	< 0.001
FBG, Median (Q1, Q3), mmol/L	5.04 (4.55, 5.81)	5.22 (4.64, 6.43)	4.95 (4.5, 5.56)	< 0.001

Screening for risk factors of CAD

The clinical variables included in this study covered basic demographics, blood biochemical parameters, auxiliary examinations, personal history, and medical history. Through univariate logistic regression analysis (Tables 3, 4), LASSO regression, and multivariate logistic regression, 13 characteristic variables were identified from the male cohort, including age, ethnicity, marital status, education level, hypertension, diabetes, diastolic blood pressure, creatinine, uric acid, total cholesterol (TC), high-density lipoprotein cholesterol (HDL-C), lipoprotein(a) (Lp(a)), and fasting blood glucose (FBG) (Figs. 1, 2). From the female cohort, 11 characteristic variables were selected, including marital status, education level, hypertension, diabetes, systolic blood pressure, creatinine, urea, uric acid, HDL-C, Lp(a), and FBG (Figs. 3, 4).

Machine learning model construction and selection

The dataset was randomly partitioned into training (70%), validation (15%), and test (15%) sets. For male participants, this resulted in 5,367 in the training set, 1150 in the validation set, and 1138 in the test set (Table 5). Similarly, female participants were divided into 3130 for the training set, 666 in the validation set, and 665 in the test set (Table 6). We developed gender-specific CAD risk prediction models using six distinct machine learning classifiers, employing tenfold cross-validation with averaged results for final evaluation. Among male participants, XGBoost achieved the highest AUC values in both the training and validation sets. It was followed, in the training set, by AdaBoost, SVM, Random Forest (RF), Multilayer Perceptron (MLP), and k-Nearest Neighbors (KNN); and in the validation set, by MLP, RF, AdaBoost, SVM, and KNN (Fig. 5). A similar performance hierarchy was observed among females, where XGBoost also ranked first. The subsequent rankings were AdaBoost, RF, MLP, KNN, and SVM in the training set, and MLP, RF, AdaBoost, SVM, and KNN in the validation set (Fig. 6). Based on comprehensive analysis of both training and validation sets, we identified XGBoost as the optimal predictive model across all classifiers, demonstrating consistently superior performance in both gender cohorts.

Table 3 Univariate logistic regression analysis for male

Variable	OR	OR_CI (Lower, Upper)	*p* value
Age	5.459	2.37,15.85	0.000
Ethnic	1.632	1.44,1.85	0.000
Marital status	2.980	2.48,3.59	0.000
Education	0.731	0.61,0.88	0.001
Current smoking	0.865	0.77,0.97	0.011
Current drinking	0.948	0.84,1.07	0.374
Hypertension	3.855	3.39,4.39	0.000
Diabetes	3.741	3.14,4.48	0.000
BMI	1.116	0.49,2.48	0.788
Heart rate	1.001	1.00,1.01	0.799
SBP	1.005	1.00,1.01	0.001
DBP	0.993	0.99,1.00	0.002
Crea	1.027	1.02,1.03	0.000
Urea	1.201	1.16,1.25	0.000
UA	1.002	1.00,1.00	0.000
TC	0.738	0.70,0.78	0.000
TG	1.019	0.98,1.06	0.330
HDL.C	0.129	0.10,0.16	0.000
LDL.C	0.726	0.68,0.77	0.000
FBG	1.186	1.15,1.22	0.000
LPa	1.001	1.00,1.00	0.000

Table 4 Univariate logistic regression analysis for female

Variable	OR	OR_CI (Lower, Upper)	*p* value
Age	1.531	0.41,9.98	0.582
Ethnic	1.273	1.08,1.50	0.004
Marital status	2.333	1.81,3.01	0.000
Education	0.626	0.52,0.75	0.000
Current smoking	1.233	0.60,2.51	0.564
Current drinking	0.945	0.52,1.67	0.849
Hypertension	6.005	5.10,7.08	0.000
Diabetes	4.173	3.35,5.23	0.000
BMI	1.329	0.69,2.72	0.412
Heart rate	0.995	0.99,1.00	0.164
SBP	1.023	1.02,1.03	0.000
DBP	0.997	0.99,1.00	0.321
Crea	0.988	0.98,0.99	0.000
Urea	1.085	1.03,1.14	0.001
UA	0.997	1.00,1.00	0.000
TC	0.994	0.93,1.06	0.862
TG	1.011	0.95,1.07	0.706
HDL.C	0.621	0.49,0.78	0.000
LDL.C	0.983	0.91,1.07	0.683
LPa	1.001	1.00,1.00	0.000
Glu	1.237	1.19,1.29	0.000

Optimization of the optimal machine learning model

XGBoost emerged as the optimal machine learning model for both gender cohorts. For male participants, the XGBoost model achieved AUROCs of 0.845 (95% CI 0.834–0.855) in the training set, 0.814 (95% CI 0.789–0.839) in the validation set, and 0.826 (95% CI 0.802–0.850) in the test set following cross-validation (Table 7, Fig. 7). In contrast, among female participants,

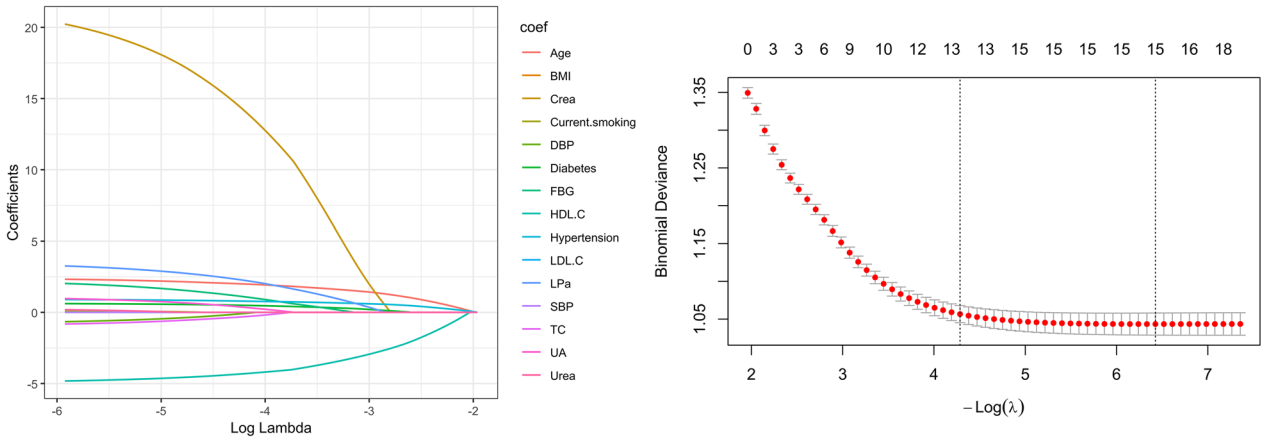


Fig.1 LASSO regression analysis of risk factors for male CAD

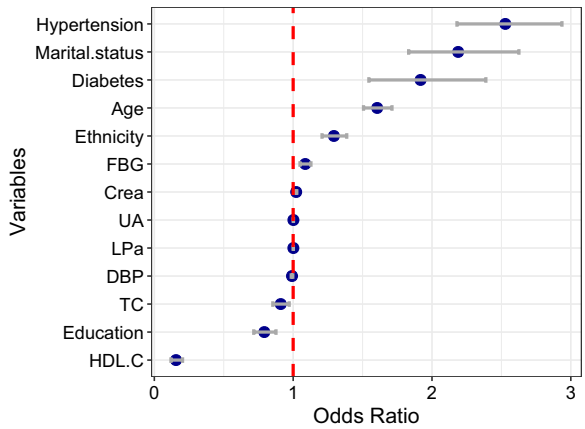


Fig.2 Forest plot of risk factors for male CAD identified by multivariate regression analysis

the XGBoost model showed AUROCs of 0.817 (95% CI 0.802–0.832) in the training set, 0.759 (95% CI 0.721–0.796) in the validation set, and 0.786 (95% CI 0.751–0.821) in the test set after cross-validation (Table 7, Fig. 8).

SHAP interpretation of the optimal machine learning model

SHAP (SHapley Additive Explanations) analysis revealed distinct gender-specific risk patterns for CAD prediction. In males, age was identified as the most dominant predictor. Advanced age (particularly > 50 years old), elevated Crea, TC, Lp(a), FBG, UA, DBP, lower HDL-C, and histories of hypertension and diabetes significantly increased the predicted risk. (Fig. 9), while features like diastolic blood pressure (DBP) and education demonstrated

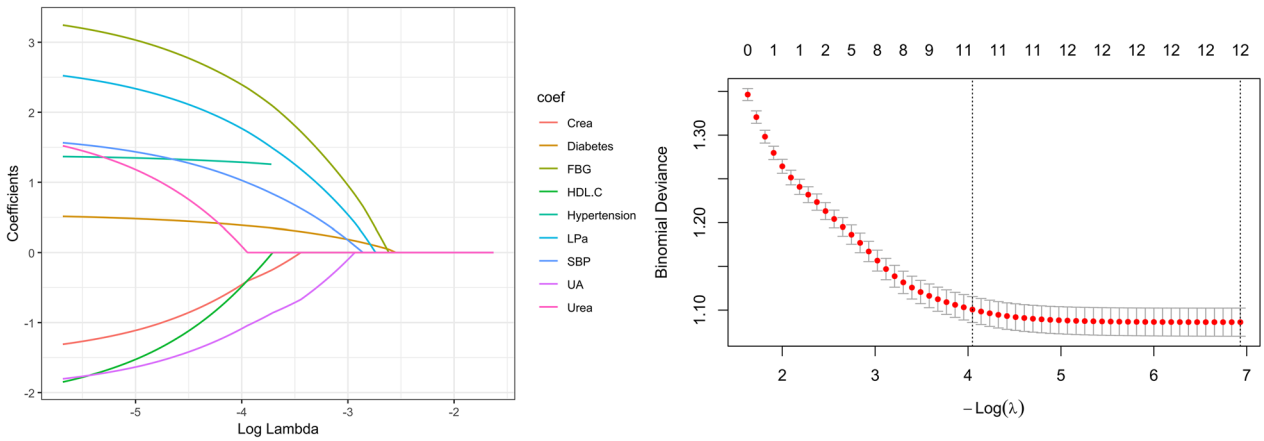


Fig.3 LASSO regression analysis of risk factors for female CAD

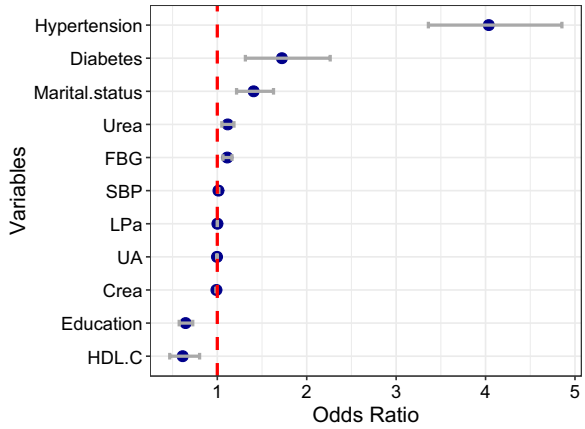


Fig.4 Forest plot of risk factors for female CAD identified by multivariate regression analysis

minimal contributions. Age exhibited a nonlinear threshold effect, with strongly positive SHAP values beyond 50 years (Fig. 11). Ethnicity analysis identified Kazakh and Hui populations with consistently positive SHAP values compared to Han and Uyghur groups with protective patterns. Conversely, higher education level and married status demonstrated protective effects (Fig. 11, Supplementary Fig. 1). Among female patients, the presence of hypertension was the most decisive factor influencing the prediction outcome. Elevated levels of UA, SBP, FBG, Lp(a), Crea, and Urea, along with lower HDL-C, a history of hypertension, and diabetes, significantly increased the model's predicted value, indicating a substantially Fig. 10 elevated risk. Conversely, being married and having a higher education level may serve as protective factors (Fig. 12, Supplementary Fig. 2). These SHAP dependence plots collectively demonstrated both the magnitude and directionality of each feature's contribution to the XGBoost model's predictions.

Model calibration and performance validation

The validation of the male prediction model demonstrated favorable clinical utility and predictive accuracy. Decision curve analysis indicated that within the 0.1–0.5 threshold range, the model's net benefit significantly outperformed both "treat all" and "treat none" strategies (peak net benefit: 0.4–0.6). The clinical impact curve showed that as the threshold increased from 0.0 to 0.2, high-risk patients decreased from 600 to 300, with corresponding reductions in true and false positives; the curve remained stable within the 0.2–0.6 threshold range, offering flexibility for clinical decision-making. The calibration curve revealed a significant improvement in predictive accuracy after calibration, which was further confirmed by Brier score analysis (the test set score decreased from 0.1670 to 0.1655, while the training and validation set scores ranged from 0.1628 to 0.1592 and from 0.1733 to 0.1721) (Fig. 13).The female model evaluation showed excellent clinical utility and calibration performance. Decision curve analysis demonstrated similar net benefit advantages within the 0.1–0.5 threshold range. The clinical impact curve exhibited a consistent trend, with high-risk patients decreasing from 600 to 300 as the threshold increased from 0.0 to 0.2. Calibration notably improved predictive accuracy, and the Brier score decreased from 0.1930 to 0.1838 in the test set (training set: 0.1765–0.1720; validation set: 0.2004–0.1937), demonstrating good generalization capability of the model (Fig. 14).The SCORE2 Asia Pacific model demonstrated an AUROC of only 0.662 (95% CI 0.640–0.684) in the male test set and 0.720 (95% CI 0.693–0.748) in the female test set within our testing cohort, showing significantly inferior discriminatory ability and calibration performance compared to our model.(Fig. 15,16).

Discussion

In this study, we developed and validated a CAD risk prediction model for multi-ethnic populations in Xinjiang using six machine learning algorithms (XGBoost, RF, MLP, SVM, KNN, and AdaBoost), with comprehensive benchmarking to identify the optimal approach. Compared to conventional statistical methods like logistic regression, machine learning offers distinct advantages for this research context. ML algorithms can directly capture nonlinear relationships and complex interactions without requiring multistep statistical analysis, while demonstrating greater efficacy in processing large-scale clinical datasets and reducing overfitting risks. Importantly, machine learning imposes fewer distributional requirements and exhibits superior adaptability to diverse medical data types [17, 18].Our analysis showed XGBoost consistently outperformed other algorithms across both gender groups, achieving AUROCs of 0.845 (95% CI 0.834–0.855) in the training set, 0.814 (95% CI 0.789–0.839) in the validation set, and 0.826 (95% CI 0.802–0.850) in the test set for male participants. For female participants, the model showed comparable performance with AUROCs of 0.817 (95% CI 0.802–0.832), 0.759 (95% CI 0.721–0.796), and 0.786 (95% CI 0.751–0.821) in the respective datasets. These results indicate that gender-specific models provide robust predictive performance for CAD risk assessment, with particularly strong discrimination accuracy observed in the male cohort. To enhance the interpretability and clinical utility of our machine learning approach, we implemented SHAP (SHapley Additive Explanations) analysis on the XGBoost model. This advanced explainable AI technique, which has gained significant recognition in medical

Table 5 Baseline characteristics of the training set, validation set and test set for male subjects

Variables	Train set		Validation set		Train set	
	CAD (*n* = 3191)	Non-CAD (*n* = 2176)	CAD (*n* = 682)	Non-CAD (*n* = 468)	CAD (*n* = 678)	Non-CAD (*n* = 460)
Age, n (%)						
18–29	5 (0)	67 (3)	3 (0)	20 (4)	2 (0)	26 (6)
30–39	132 (4)	324 (15)	25 (4)	59 (13)	26 (4)	69 (15)
40–49	649 (20)	693 (32)	136 (20)	155 (33)	135 (20)	130 (28)
50–59	1191 (37)	643 (30)	263 (39)	129 (28)	274 (40)	112 (24)
60–69	845 (26)	328 (15)	175 (26)	74 (16)	181 (27)	96 (21)
≥ 70	369 (12)	121 (6)	80 (12)	31 (7)	60 (9)	27 (6)
Ethnic, *n* (%)						
Han	1177 (37)	1118 (51)	229 (34)	253 (54)	256 (38)	243 (53)
Uyghur	1244 (39)	724 (33)	284 (42)	157 (34)	270 (40)	149 (32)
Kazakh	396 (12)	167 (8)	91 (13)	39 (8)	78 (12)	34 (7)
Hui	374 (12)	167 (8)	78 (11)	19 (4)	74 (11)	34 (7)
Marital status, *n* (%)						
Married	2523 (79)	1996 (92)	528 (77)	427 (91)	547 (81)	428 (93)
Never married	599 (19)	159 (7)	138 (20)	34 (7)	123 (18)	25 (5)
Divorced or Widowed	69 (2)	21 (1)	16 (2)	7 (1)	8 (1)	7 (2)
Education levels, *n* (%)						
Elementary	496 (16)	196 (9)	109 (16)	41 (9)	102 (15)	44 (10)
Middle	1605 (50)	868 (40)	349 (51)	192 (41)	346 (51)	185 (40)
Higher	1090 (34)	1112 (51)	224 (33)	235 (50)	230 (34)	231 (50)
Current smoking, *n* (%)						
Yes	1152 (36)	860 (40)	241 (35)	177 (38)	243 (36)	166 (36)
No	2039 (64)	1316 (60)	441 (65)	291 (62)	435 (64)	294 (64)
Current drinking, *n* (%)						
Yes	987 (31)	698 (32)	227 (33)	135 (29)	213 (31)	138 (30)
No	2204 (69)	1478 (68)	455 (67)	333 (71)	465 (69)	322 (70)
Hypertension, *n* (%)						
Yes	1456 (46)	389 (18)	307 (45)	86 (18)	310 (46)	71 (15)
No	1735 (54)	1787 (82)	375 (55)	382 (82)	368 (54)	389 (85)
Diabetes, *n* (%)						
Yes	757 (24)	167 (8)	147 (22)	39 (8)	159 (23)	45 (10)
No	2434 (76)	2009 (92)	535 (78)	429 (92)	519 (77)	415 (90)
BMI, *n* (%),kg/m2						
< 18.5	14 (0)	11 (1)	3 (0)	2 (0)	3 (0)	2 (0)
18.5–23.9	632 (20)	445 (20)	134 (20)	121 (26)	122 (18)	100 (22)
24.0–27.9	1403 (44)	1024 (47)	311 (46)	192 (41)	282 (42)	215 (47)
≥ 28	1142 (36)	696 (32)	234 (34)	153 (33)	271 (40)	143 (31)
Heart rate, Median (Q1, Q3), beats/min	78 (71, 85)	78 (72, 84)	78 (72, 85)	78 (71, 84)	77.5 (70, 86)	77 (71, 82)
SBP, Median (Q1, Q3), mmHg	125 (115, 139)	123 (115, 136)	125 (115, 139)	121 (114, 137)	126 (114.25, 139)	122 (112, 135)
DBP, Median (Q1, Q3), mmHg	77 (70, 84)	79 (70, 85)	76 (70, 84.75)	76 (69.75, 84)	78 (70, 85)	79 (70, 86)
Crea, Median (Q1, Q3), μmol/L	77 (67, 87.82)	70 (60, 81)	76 (66.3, 87.96)	72 (61.46, 82.05)	77 (67, 87.74)	71 (60.03, 83)

Table 5 (continued)

Variables	Train set		Validation set		Train set	
	CAD (*n* = 3191)	Non-CAD (*n* = 2176)	CAD (*n* = 682)	Non-CAD (*n* = 468)	CAD (*n* = 678)	Non-CAD (*n* = 460)
Urea, Median (Q1, Q3), mmol/L	5.5 (4.6, 6.6)	5.2 (4.3, 6.1)	5.5 (4.6, 6.5)	5.12 (4.39, 6.2)	5.5 (4.7, 6.51)	5.1 (4.4, 6.03)
UA, Median (Q1, Q3), μmol/L	338 (287.75, 395)	319 (262.6, 382.22)	338 (282.18, 392.98)	319 (256.98, 377.48)	332.9 (286.45, 392.75)	324.7 (266.85, 386.25)
TC, Median (Q1, Q3), mmol/L	3.8 (3.1, 4.7)	4.26 (3.67, 4.95)	3.76 (3.06, 4.65)	4.27 (3.57, 4.91)	3.8 (3.09, 4.7)	4.23 (3.63, 4.93)
TG, Median (Q1, Q3), mmol/L	1.48 (1.04, 2.12)	1.42 (0.98, 2.07)	1.46 (1, 2.03)	1.39 (0.97, 1.98)	1.46 (1.04, 2.14)	1.39 (0.99, 2.05)
HDL-C, Median (Q1, Q3), mmol/L	0.94 (0.81, 1.11)	1.09 (0.91, 1.3)	0.95 (0.82, 1.11)	1.09 (0.9, 1.31)	0.94 (0.81, 1.1)	1.09 (0.94, 1.32)
LDL-C, Median (Q1, Q3), mmol/L	2.49 (1.89, 3.2)	2.83 (2.32, 3.38)	2.45 (1.9, 3.16)	2.8 (2.29, 3.38)	2.47 (1.89, 3.2)	2.78 (2.25, 3.34)
LP(a), Median (Q1, Q3), mg/L	5.26 (4.55, 6.48)	4.99 (4.5, 5.63)	5.1 (4.47, 6.38)	4.93 (4.51, 5.54)	5.14 (4.46, 6.33)	5.02 (4.58, 5.59)
FBG, Median (Q1, Q3), mmol/L	142 (66.7, 293.52)	105.8 (49.7,210.06)	150.99 (67.11, 299.09)	99.09 (48.62, 206.09)	127.84 (64.47, 275.32)	93.9 (47.93, 200.19)

research, particularly for cardiovascular disease prediction [19], provides robust quantitative assessment of feature importance. The SHAP framework enables precise evaluation of each variable's contribution to model predictions, while SHAP decision plots offer intuitive visualization of individualized risk assessment processes within our XGBoost algorithm. Our findings identified advanced age, elevated Crea, TC, Lp(a), FBG, UA, DBP, lower HDL-C, hypertension, and diabetes as significant risk factors for CAD in males. Conversely, higher education and married status emerged as protective factors. Ethnically, Kazakh and Hui populations were associated with elevated risk, whereas Han and Uygur ethnicity demonstrated a protective trend. Among female subjects, the presence of hypertension was the most decisive factor influencing the prediction outcome. Elevated levels of UA, SBP, FBG, Lp(a), Crea, and Urea, along with lower HDL-C, a history of hypertension, and diabetes, significantly increased the model's predicted value, indicating a substantially elevated risk. Conversely, being married and having a higher education level may serve as protective factors. The prediction model demonstrated favorable clinical utility and accuracy in both cohorts, with calibration significantly enhancing predictive performance. Compared to the SCORE2 Asia Pacific risk model, our model exhibited superior discriminatory ability and improved calibration performance.

Since the first CAD risk equation was developed by the Heart Disease Study in 1976, several CAD risk assessment tools have been established, guiding public health and clinical practice across different populations, including QRISK and PCE [8–10]. However, the PCE model performed poorly when applied to East Asian populations [20]. Since these equations were derived from Western population samples, their generalizability to other ethnic groups is limited [10, 21]. Wu et al. developed a 10-year risk prediction model for ischemic CVD (including ischemic stroke and coronary events) using derivation and validation cohorts of 9903 and 17,329 subjects, respectively. However, both prediction models were derived at least a decade ago. Over the past ten years, the prevalence of cardiovascular diseases and their risk factors have evolved. Risk prediction equations developed from recent data would better generalize to contemporary populations and more accurately identify high-risk individuals with appropriate predicted risk probabilities [22]. To address this gap, Yang et al. developed the China-PAR model to predict the 10-year risk of ASCVD in the Chinese population. The sex-specific equations demonstrated C statistics of 0.794 (95% CI 0.775–0.814) for men and 0.811 (95% CI 0.787–0.835) for women [11]. However, their study primarily included Han Chinese individuals, while China is a vast country with diverse ethnic groups. Significant variations exist across different regions and ethnicities in terms of genetic background [12, 13],life style, and dietary habits. Moreover, Yang et al.'s findings revealed substantial regional disparities in the incidence of cardiovascular disease subtypes, suggesting that a universal model may not be directly applicable to different regions of China [14].Jiang et al. externally validated three ASCVD risk prediction equations in a rural population from Xinjiang. The results showed that for male patients, the C statistics (95% CI) for the PCE, PAR, and FRS models were 0.727 (0.689–0.766), 0.727

Table 6 Baseline characteristics of the training set, validation set and test set for female subjects

Variables	Train set		Validation set		Test set	
	CAD (*n* = 1255)	Non-CAD (*n* = 1875)	CAD (*n* = 266)	Non-CAD (*n* = 400)	CAD (*n* = 266)	Non-CAD (*n* = 399)
Age, *n* (%)						
18–29	2 (0)	29 (2)	0 (0.0)	3 (0.8)	0 (0.0)	6 (1.5)
30–39	17 (1)	161 (9)	4 (1.5)	30 (7.5)	2 (0.8)	25 (6.3)
40–49	95 (8)	441 (24)	16 (6.0)	100 (25.0)	26 (9.8)	107 (26.8)
50–59	424 (34)	725 (39)	91 (34.2)	146 (36.5)	86 (32.3)	149 (37.3)
60–69	487 (39)	405 (22)	105 (39.5)	92 (23.0)	103 (38.7)	78 (19.5)
≥ 70	230 (18)	114 (6)	50 (18.8)	29 (7.2)	49 (18.4)	34 (8.5)
Ethnic, *n* (%)						
Han	514 (41)	902 (48)	102 (38.3)	179 (44.8)	115 (43.2)	177 (44.4)
Uyghur	444 (35)	612 (33)	90 (33.8)	138 (34.5)	89 (33.5)	136 (34.1)
Kazakh	137 (11)	188 (10)	33 (12.4)	43 (10.8)	28 (10.5)	49 (12.3)
Hui	160 (13)	173 (9)	41 (15.4)	40 (10.0)	34 (12.8)	37 (9.3)
Marital status, *n* (%)						
Married	961 (77)	1674 (89)	214 (80.5)	357 (89.2)	216 (81.2)	364 (91.2)
Never married	154 (12)	115 (6)	28 (10.5)	22 (5.5)	27 (10.2)	17 (4.3)
Divorced or Widowed	140 (11)	86 (5)	24 (9.0)	21 (5.2)	23 (8.6)	18 (4.5)
Education, *n* (%)						
Elementary	405 (32)	333 (18)	95 (35.7)	79 (19.8)	79 (29.7)	80 (20.1)
Middle	631 (50)	829 (44)	124 (46.6)	180 (45.0)	138 (51.9)	183 (45.9)
Higher	219 (17)	713 (38)	47 (17.7)	141 (35.2)	49 (18.4)	136 (34.1)
Current smoking, *n* (%)						
Yes	14 (1)	17 (1)	4 (1.5)	6 (1.5)	5 (1.9)	3 (0.8)
No	1241 (99)	1858 (99)	262 (98.5)	394 (98.5)	261 (98.1)	396 (99.2)
Current drinking, *n* (%)						
Yes	19 (2)	30 (2)	2 (0.8)	8 (2.0)	7 (2.6)	2 (0.5)
No	1236 (98)	1845 (98)	264 (99.2)	392 (98.0)	259 (97.4)	397 (99.5)
Hypertension, *n* (%)						
Yes	696 (55)	322 (17)	135 (50.8)	73 (18.2)	150 (56.4)	88 (22.1)
No	559 (45)	1553 (83)	131 (49.2)	327 (81.8)	116 (43.6)	311 (77.9)
Diabetes, *n* (%)						
Yes	292 (23)	127 (7)	52 (19.5)	33 (8.2)	61 (22.9)	22 (5.5)
No	963 (77)	1748 (93)	214 (80.5)	367 (91.8)	205 (77.1)	377 (94.5)
BMI, *n* (%),kg/m2						
< 18.5	12 (1)	31 (2)	2 (0.8)	2 (0.5)	4 (1.5)	8 (2.0)
18.5–23.9	358 (29)	696 (37)	82 (30.8)	151 (37.8)	69 (25.9)	148 (37.1)
24.0–27.9	497 (40)	712 (38)	104 (39.1)	141 (35.2)	119 (44.7)	149 (37.3)
≥ 28	388 (31)	436 (23)	78 (29.3)	106 (26.5)	74 (27.8)	94 (23.6)
Heart rate, Median (Q1, Q3), beats/min	77 (70.5, 84)	78 (71, 84)	76.00 [68.25, 83.00]	78.00 [72.00, 85.00]	77.00 (71.25, 83.00)	78.00 (72.00, 84.00)
SBP, Median (Q1, Q3), mmHg	129 (119.5, 143)	122 (112, 134)	132.00 [119.25, 145.00]	121.00 [112.00, 134.00]	130.00 (119.00, 144.00)	122.00(112.00, 134.00)
DBP, Median (Q1, Q3), mmHg	76 (69, 82)	76 (70, 82)	76.00 [68.00, 82.00]	75.00 [69.00, 81.00]	75.00 (67.00, 83.75)	76.00(70.00, 83.00)
Crea, Median (Q1, Q3), μmol/L	61 (52, 70.55)	63.32 (53.81, 74)	60.56 [51.06, 69.97]	63.62 [55.00, 74.70]	60.00 (52.00, 69.95)	63.56(54.38, 72.70)

Table 6 (continued)

Variables	Train set		Validation set		Test set	
	CAD (*n* = 1255)	Non-CAD (*n* = 1875)	CAD (*n* = 266)	Non-CAD (*n* = 400)	CAD (*n* = 266)	Non-CAD (*n* = 399)
Urea, Median (Q1, Q3), mmol/L	5 (4.3, 6.1)	4.91 (4.1, 5.9)	5.04 [4.10, 6.01]	4.80 [4.00, 5.67]	5.10 (4.23, 6.00)	5.00(4.20, 5.90)
UA, Median (Q1, Q3), μmol/L	269.2 (229, 317.54)	286.9 (234, 344.3)	268.75 [225.32, 331.75]	272.20 [224.00, 330.38]	264.65 (230.25, 310.95)	287.34(240.00, 341.85)
TC, Median (Q1, Q3), mmol/L	4.22 (3.49, 5.08)	4.33 (3.68, 4.98)	4.28 [3.49, 5.23]	4.32 [3.67, 4.95]	4.30 (3.48, 4.96)	4.29(3.64, 5.00)
TG, Median (Q1, Q3), mmol/L	1.39 (1.04, 1.97)	1.32 (0.94, 1.9)	1.36 [1.01, 2.00]	1.35 [0.95, 1.89]	1.41 (1.02, 1.97)	1.31(0.96, 1.81)
HDL-C, Median (Q1, Q3), mmol/L	1.12 (0.96, 1.32)	1.15 (0.96, 1.4)	1.15 [0.95, 1.31]	1.18 [1.00, 1.38]	1.09(0.95, 1.31)	1.14(0.98, 1.38)
LDL-C, Median (Q1, Q3), mmol/L	2.7 (2.13, 3.44)	2.84 (2.32, 3.39)	2.82 [2.14, 3.56]	2.86 [2.33, 3.34]	2.80(2.20, 3.41)	2.83(2.31, 3.36)
LP(a), Median (Q1, Q3), mg/L	150.45 (73.95, 314.5)	112 (55.64, 223.95)	153.77 [67.85, 298.85]	111.06 [56.35, 221.10]	148.34(70.91, 336.03)	119.10(57.78, 215.90)
FBG, Median (Q1, Q3), mmol/L	5.23 (4.66, 6.44)	4.97 (4.49, 5.56)	5.15 [4.53, 6.22]	4.92 [4.59, 5.56]	5.32(4.64, 6.53)	4.91(4.44, 5.56)

(0.684–0.770), and 0.740(0.703–0.777), respectively. For female patients, the corresponding values were 0.738 (0.703–0.773), 0.731 (0.696–0.766), and 0.761 (0.728–0.794) [23].Qian et al.'s study on ASCVD prediction models in rural populations of Xinjiang demonstrated that after constructing sex-specific models using different methods, the Random Survival Forest (RSF) model achieved the highest AUROC value for men (0.791, 95% CI 0.767–0.813), while for women, the Lasso–Cox model (0.759, 95% CI 0.733–0.783) and RSF model (0.759, 95% CI 0.734–0.783) yielded equal AUROC values. In this population, the China-PAR model exhibited moderate discriminative ability, with AUROC values of 0.757 (95% CI 0.732–0.781) for men and 0.756 (95% CI 0.730–0.780) for women [24]. In summary, when applied to CAD risk prediction across diverse regions and ethnic groups, the predictive performance of the China-PAR model declines. This substantial variation in predictive accuracy may be attributed to racial heterogeneity, differing risk profiles in CVD burden, and disparities in the treatment and control rates of risk factors (e.g., hypertension, hyperlipidemia, and diabetes) [11, 27]. Xinjiang Uygur Autonomous Region is a large multi-ethnic area in northwest China, predominantly inhabited by Han, Uygur, Kazakh, and Hui populations. Our model incorporated these four major ethnic groups in Xinjiang along with key risk factors including Lp(a), TC, HDL-C, Crea, and FBG. Notably, previous Chinese prediction models had not included ethnicity or lipoprotein(a) Lp(a) as variables. The inclusion of these parameters has partially mitigated the performance decline caused by ethnic heterogeneity, enhanced our model's discriminative ability, and improved its predictive validity for Xinjiang's multi-ethnic population. In our optimal model, male participants demonstrated AUROC values of 0.845 (95% CI 0.834–0.855) in the training set, 0.814 (95% CI 0.789–0.839) in the validation set, and 0.826 (95% CI 0.802–0.850) in the test set—all superior to the China-PAR model's predictive performance (*C* statistics: 0.794, 95% CI 0.775–0.814 for males) [11]. This advantage was further confirmed through external validation of the China-PAR model in rural Xinjiang populations, establishing our model's better predictive capability for Xinjiang's diverse ethnic groups. In our model, the optimal model for female participants demonstrated AUROC values of 0.817 (95% CI 0.802–0.832) in the training set, 0.759 (95% CI 0.721–0.796) in the validation set, and 0.786 (95% CI 0.751–0.821) in the test set. While these model performance metrics were slightly lower than those of the China-PAR model (AUROC: 0.811, 95% CI 0.787–0.835 for females), external validation of the China-PAR model in rural Xinjiang populations revealed that our model exhibited superior predictive performance for Xinjiang's multi-ethnic populations. This finding was further corroborated by studies conducted by Qian et al. [23, 24].

To address the decline in predictive performance due to population heterogeneity, the SCORE2 Asia Pacific Writing Group systematically recalibrated the sex-specific and competing risk-adjusted SCORE2 algorithm to reflect cardiovascular disease incidence rates across four risk regions in the Asia Pacific area [25, 26]. The recalibrated model demonstrated moderate to good discrimination across all regions, with an overall pooled C index of 0.710 (95% CI 0.677–0.744). Although this model is a

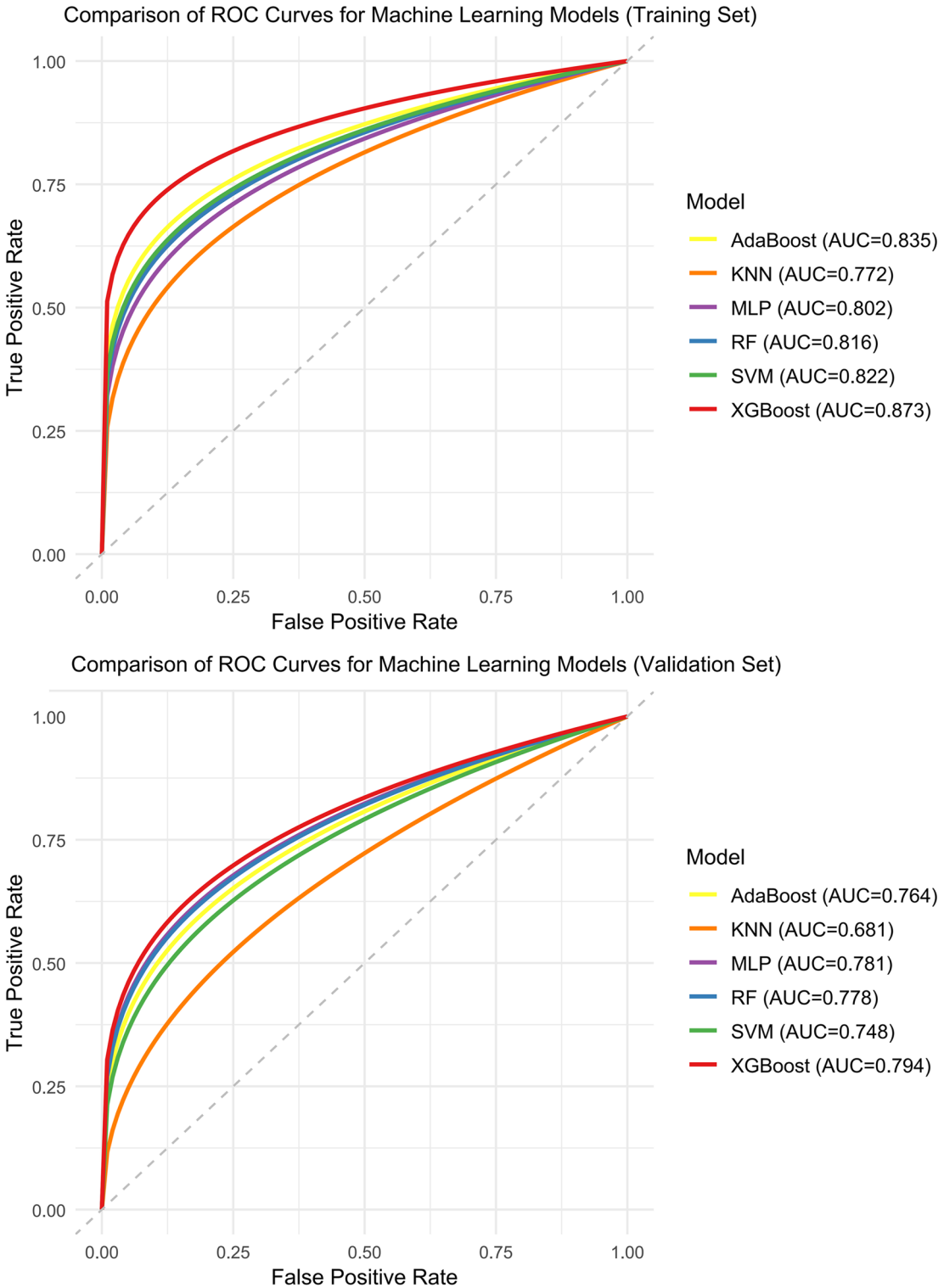
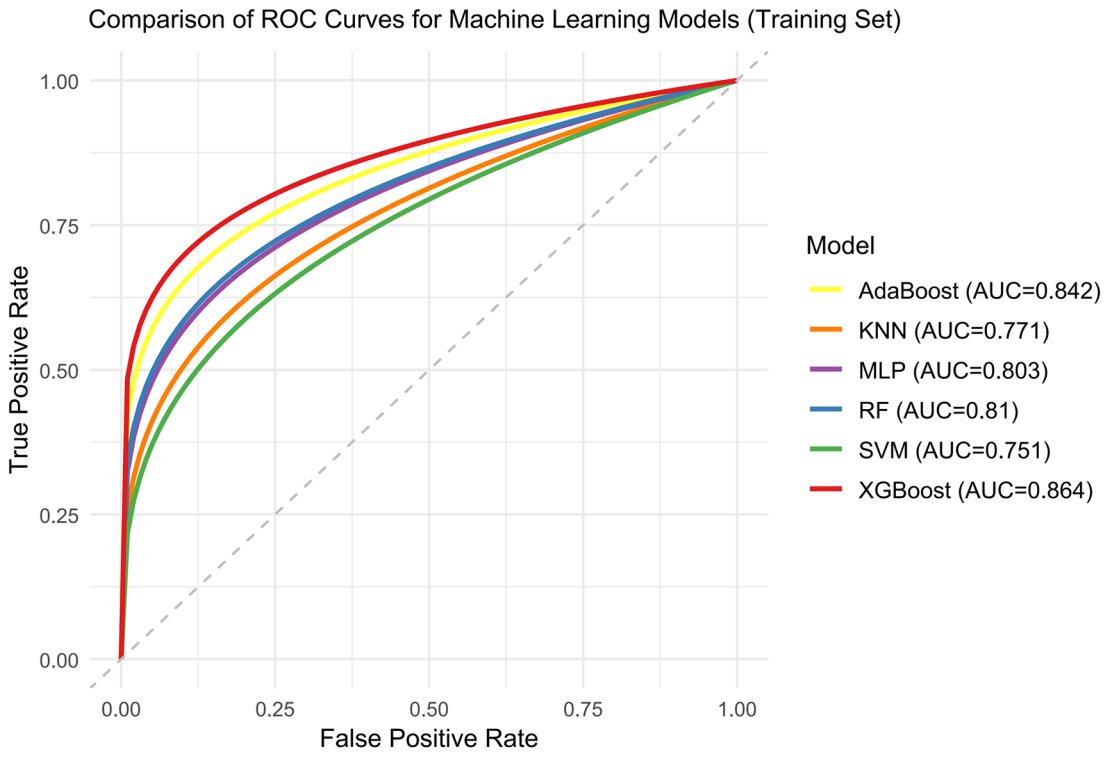


Fig.5 Comparison of machine learning classifiers for predicting CAD risk in male subjects



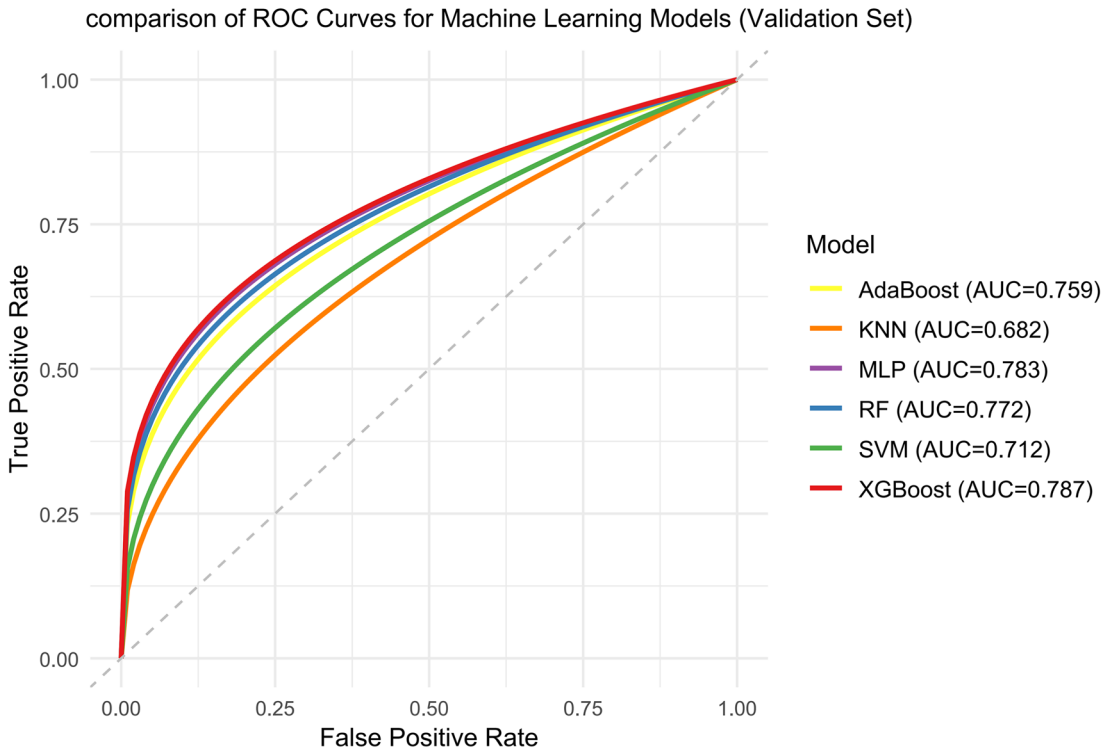


Fig.6 Comparison of machine learning classifiers for predicting CAD risk in female subjects

Table 7 Prediction performance of the best machine learning model after tenfold cross-validation

Performance	XGBoost model					
	Male			Female		
	Train set	Validation set	Test set	Train set	Validation set	Test set
AUROC(95%CI)	0.845 (0.834, 0.855)	0.814 (0.789, 0.839)	0.826 (0.802, 0.850)	0.817 (0.802, 0.832)	0.759 (0.721, 0.796)	0.786(0.751,0.821)
Accuracy	0.762	0.755	0.742	0.745	0.694	0.713
Sensitivity	0.760	0.890	0.800	0.695	0.673	0.763
Specificity	0.765	0.553	0.652	0.778	0.708	0.679

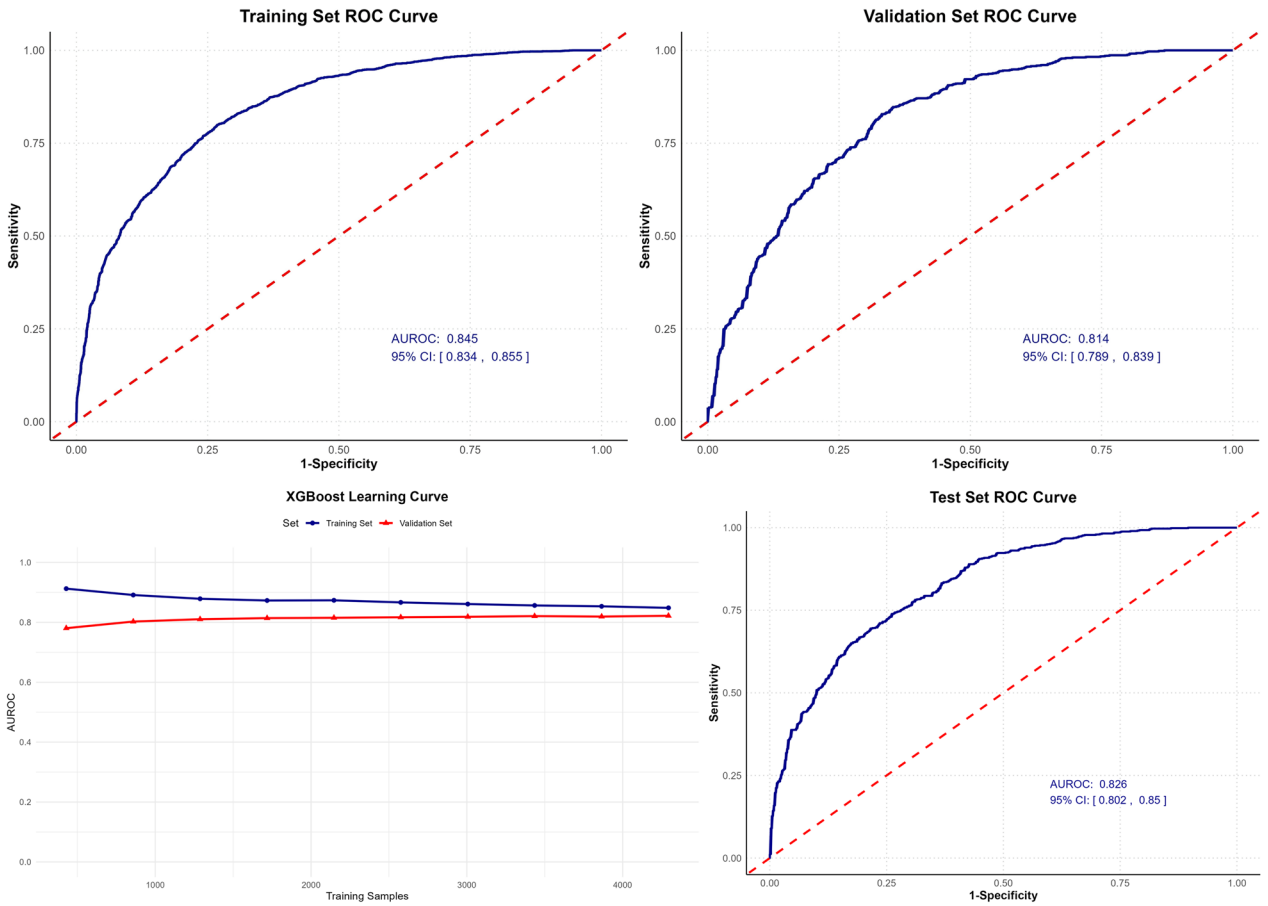


Fig.7 Hyperparameter tuning process of the prediction model for male subjects

carefully calibrated 10-year cardiovascular risk prediction tool, while our study develops a diagnostic model—representing a fundamental difference in nature and purpose—we nevertheless evaluated the SCORE2 Asia Pacific model as a diagnostic tool within our dataset to further substantiate the necessity of our own model. The results showed that the SCORE2 Asia Pacific model performed inferior to our newly developed model within our specific population and data context: in terms of discrimination, it achieved an AUROC of only 0.662 (95% CI 0.640–0.684) in the male test set and 0.720 (95% CI 0.693-0.748) in the female test set; the calibration curve also indicated significant mis-calibration. This phenomenon may stem from two main reasons: first, the Asia Pacific calibrated version of the model did not include population data from Xinjiang; second, there is an inherent limitation in applying a prospective risk prediction tool to a cross-sectional diagnostic scenario. These findings further underscore the necessity of developing localized models tailored to specific populations and clinical

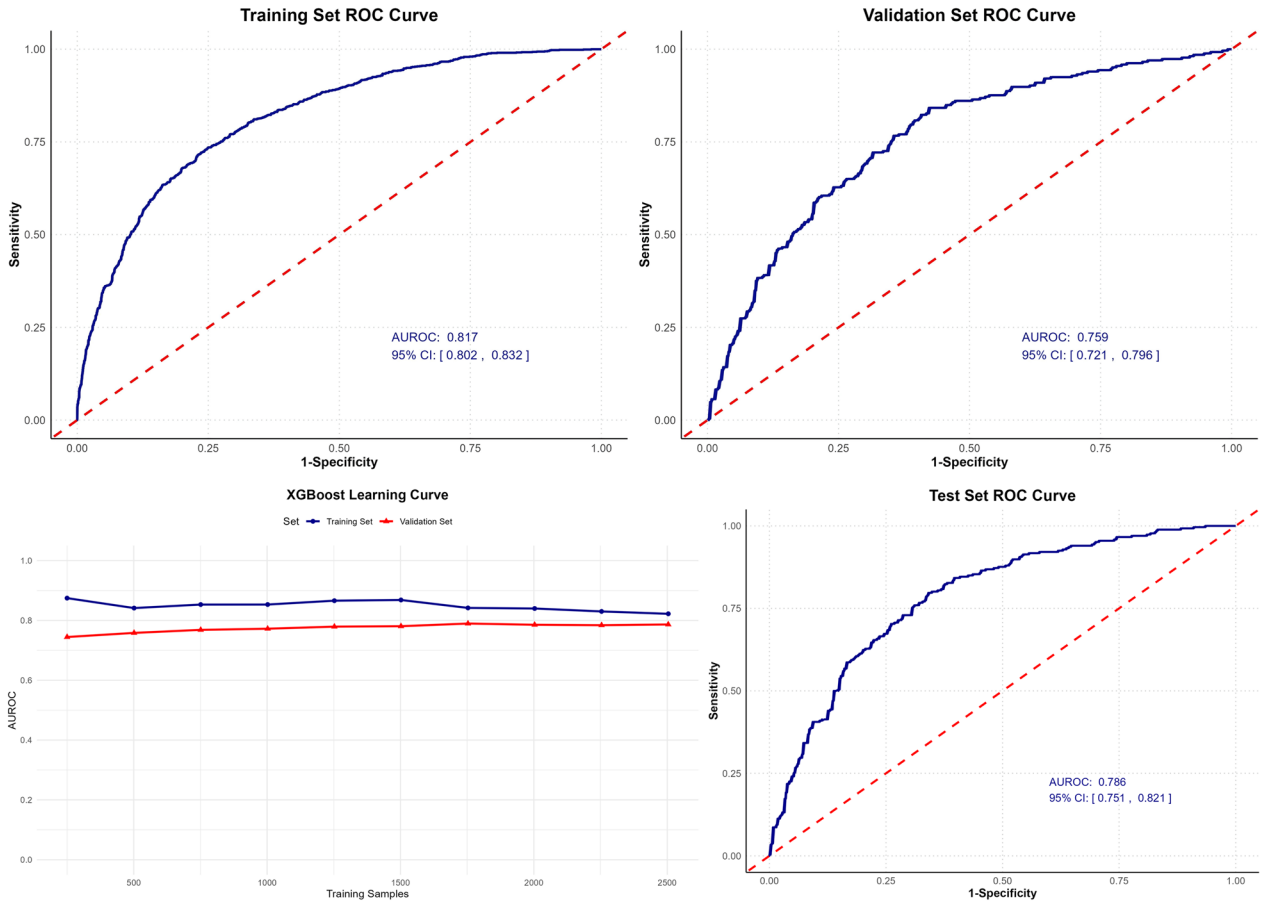


Fig.8 Hyperparameter tuning process of the prediction model for female subjects

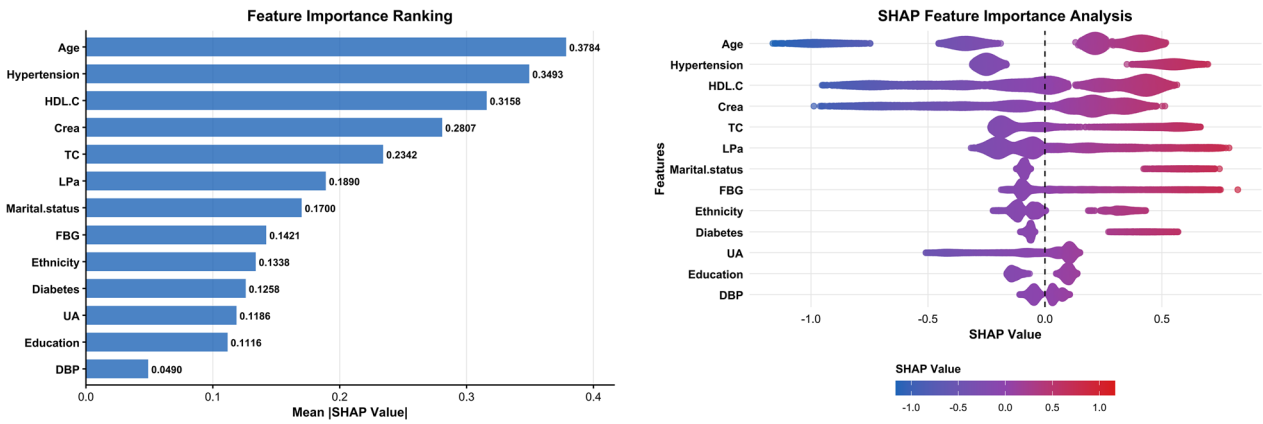


Fig.9 Feature importance plot and SHAP summary plot for male subjects

contexts. Especially for the Xinjiang region, where medical resources are relatively limited, building models suitable for auxiliary diagnosis—rather than long-term risk prediction—holds significant practical importance and clinical value.

In our study, the SHAP dependence plot for the "ethnicity" feature in male participants revealed that Kazakh and Hui ethnicities tended to significantly increase the model's predicted risk, whereas Han and Uygur ethnicities showed the opposite trend. Xinjiang Uygur Autonomous

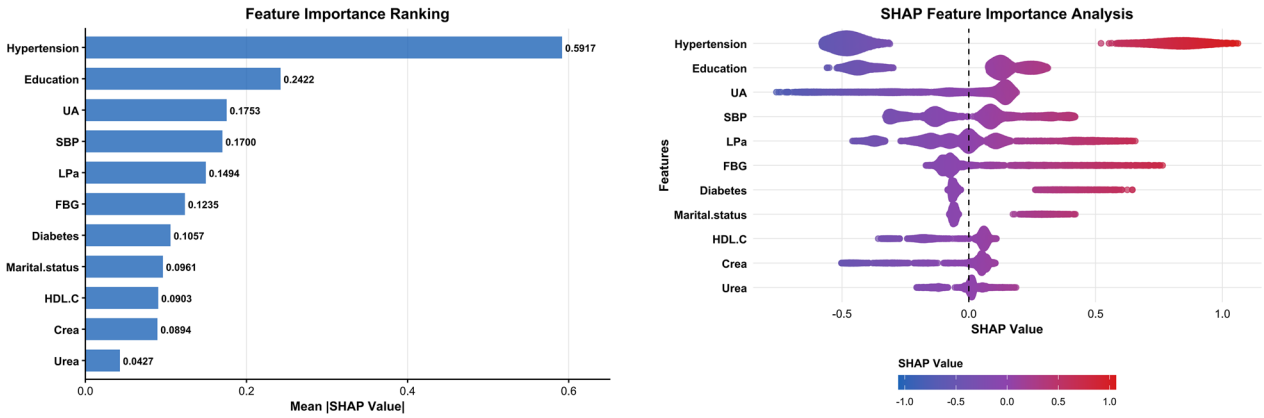


Fig.10 SHAP dependence plot for male subjects

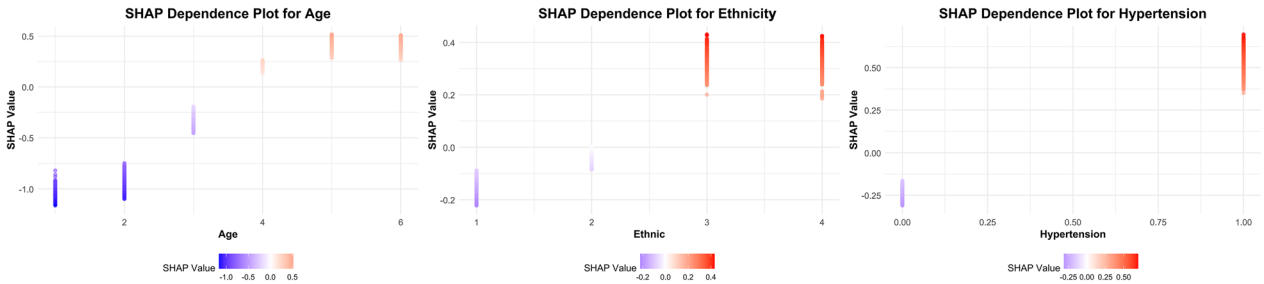


Fig.11 Feature importance plot and SHAP summary plot for female subjects

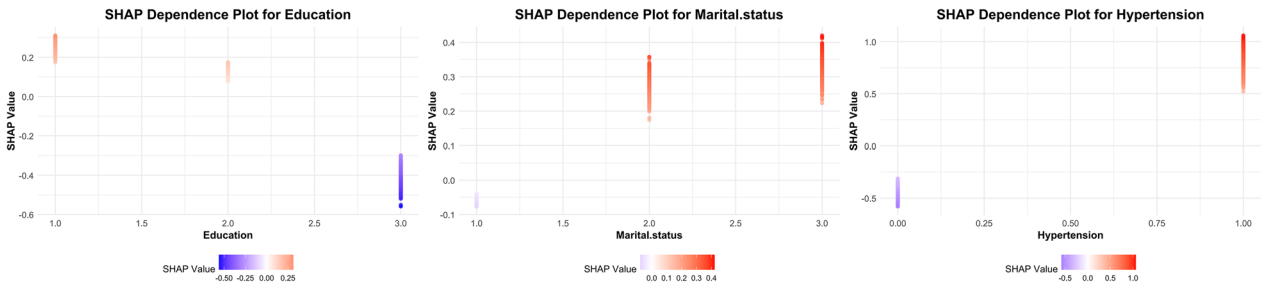


Fig.12 SHAP dependence plot for female subjects

Region, located in northwest China, is a vast multi-ethnic area predominantly inhabited by Han, Uygur, Kazakh, and Hui populations [28]. Genetic studies indicate that the Kazakh population exhibits four major ancestral components originating from East Asia, Siberia, western Eurasia, and South Asia. With over 1.25 million Kazakh residents in Xinjiang, previous research has demonstrated that this population has the highest hypertension prevalence among all Chinese ethnic minorities [29]. An epidemiological investigation of CVD risk factors among Xinjiang's ethnic groups reported that compared to Han participants, Uygur participants had adjusted odds ratios of 1.16, 1.29, and 1.55 for having 1, 2, and ≥ 3 CVD risk factors respectively, while Kazakh participants showed significantly higher ratios of 1.72, 2.75, and 4.03 [27]. This suggests Kazakh individuals accumulate substantially more CVD risk factors than Uygur and Han populations. Supporting evidence comes from Wang et al.'s study demonstrating dyslipidemia prevalence of 52.2%, 49.8%, and 42.1% in Kazakh, Uygur, and Han CAD patients respectively [30]. Given that dyslipidemia is a major CVD risk factor, these findings indirectly validate our results

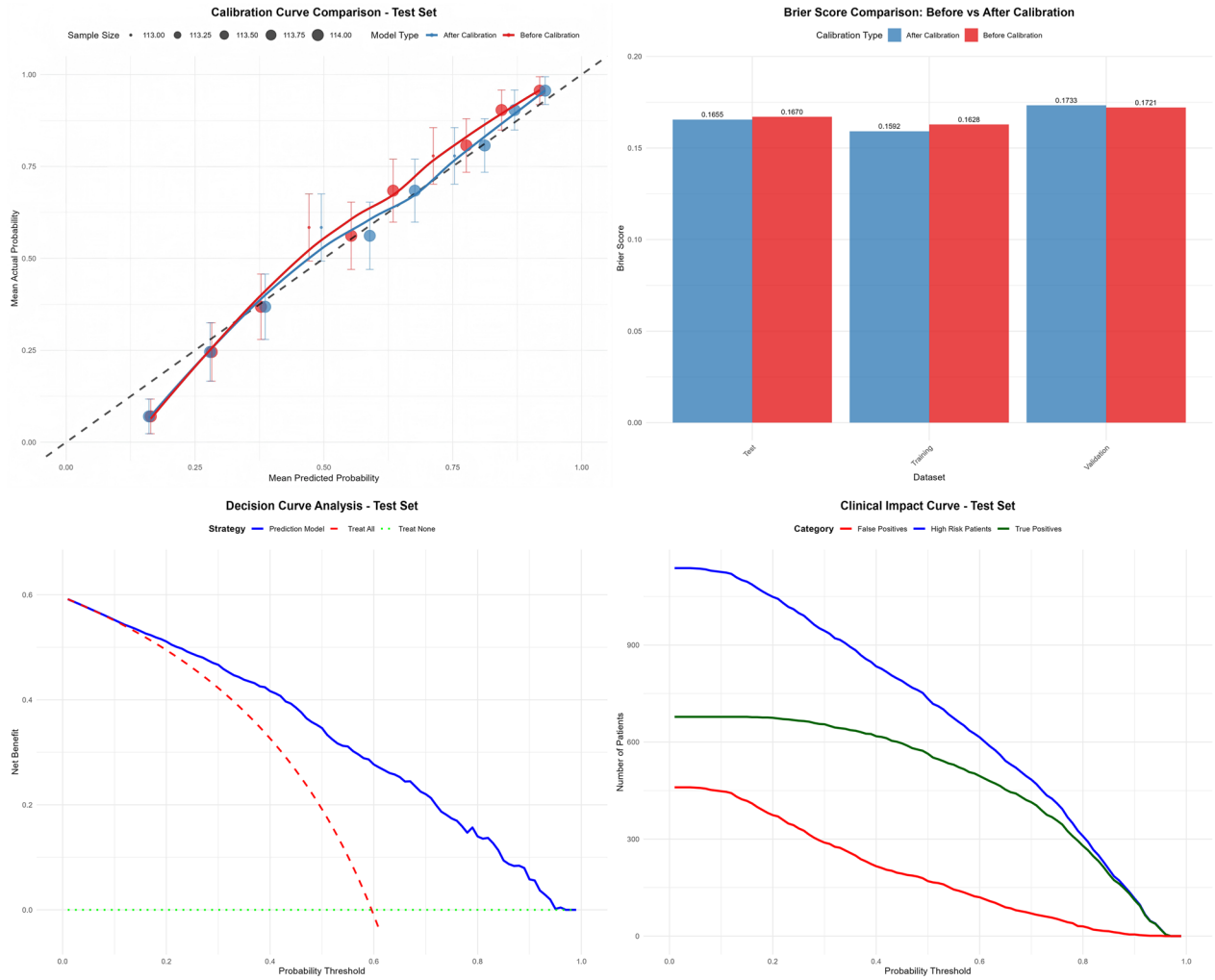


Fig.13 Multi-dimensional Assessment of Model Performance in male Subjects

[31]. Our study further highlights hypertension as a significant susceptibility factor for CAD across all genders, which may particularly contribute to the elevated CAD susceptibility observed in Kazakh patients. The Hui people are the second-largest ethnic minority among China's 55 officially recognized minority groups. Interestingly, although the Hui and Han Chinese share very similar physical appearances, they are genetically distinct, which may be attributed to the Hui's Western ancestry [32]. Among the Hui population living in Xinjiang, 6.7% of their ancestry can be traced to Western populations [33]. Previous research has shown that the prevalence of metabolic syndrome (MS) is higher among the Hui population compared to the Han population (Hui: 13.7%, Han: 10.3%) [34], a finding also supported by Qin et al. (Hui: 22.82%, Han: 19.80%) [35]. These differences may be due to a combination of genetic and environmental factors, including diet and life style. The Hui people have a long-standing preference for animal fats (such as butter and lamb fat) and fried foods. These life style differences may be one of the key factors contributing to the disparities in MS prevalence between the Hui and other ethnic groups [34, 35]. Additionally, some studies have indicated that the Hui population exhibits a higher prevalence of cardiovascular risk factors compared to the Han majority and most other ethnic minorities in China [36], further validating our findings. There are many confounding factors and random biases among different ethnic groups, such as population genetic structure, life style habits, and dietary patterns, all of which collectively influence the development of cardiovascular diseases. However, due to the limited sample size of Kazakh and Hui participants in this study, further validation with larger cohorts is needed to confirm these conclusions.

The predictive model developed in this study demonstrated strong discriminatory ability in internal

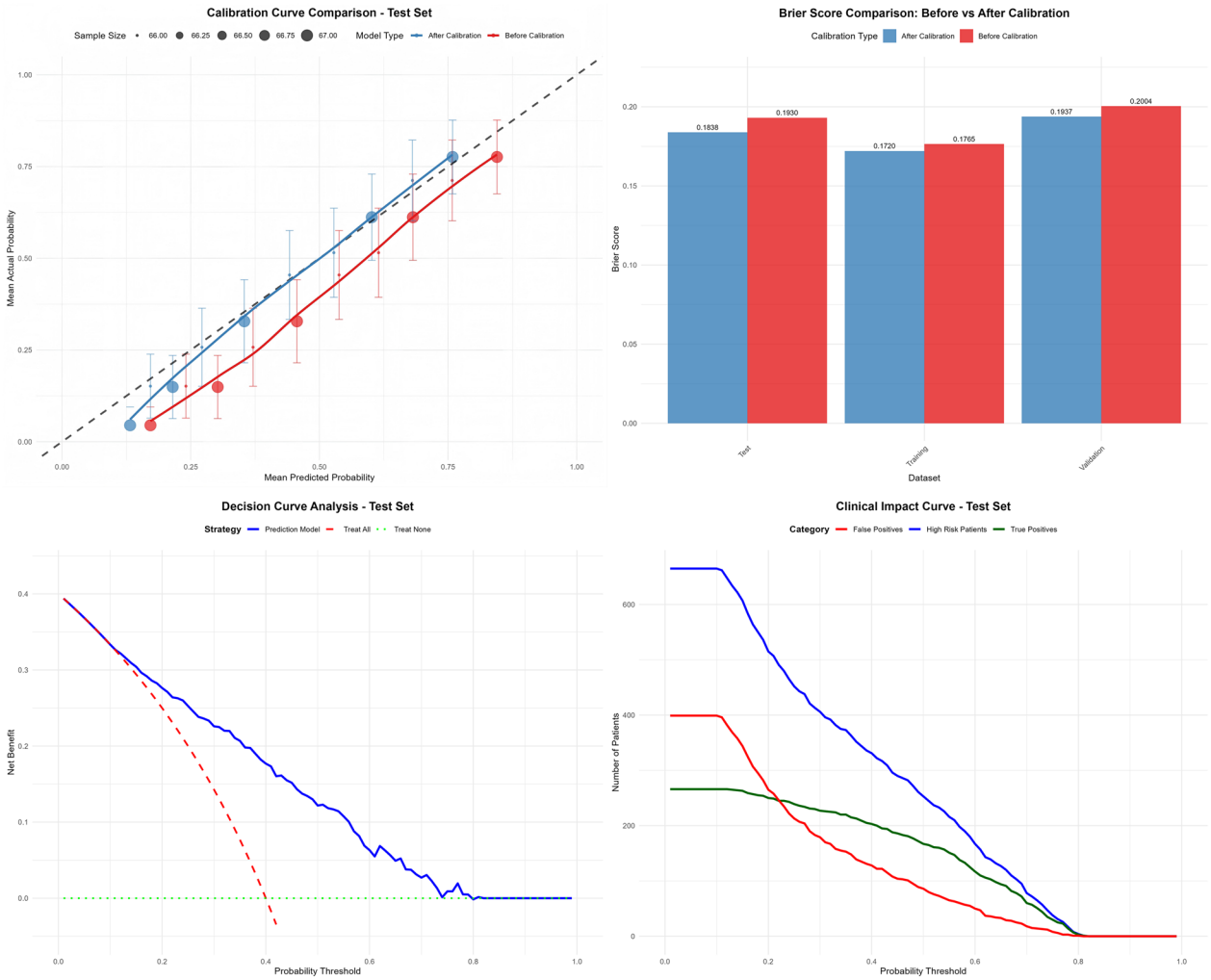


Fig.14 Multi-dimensional Assessment of Model Performance in female Subjects

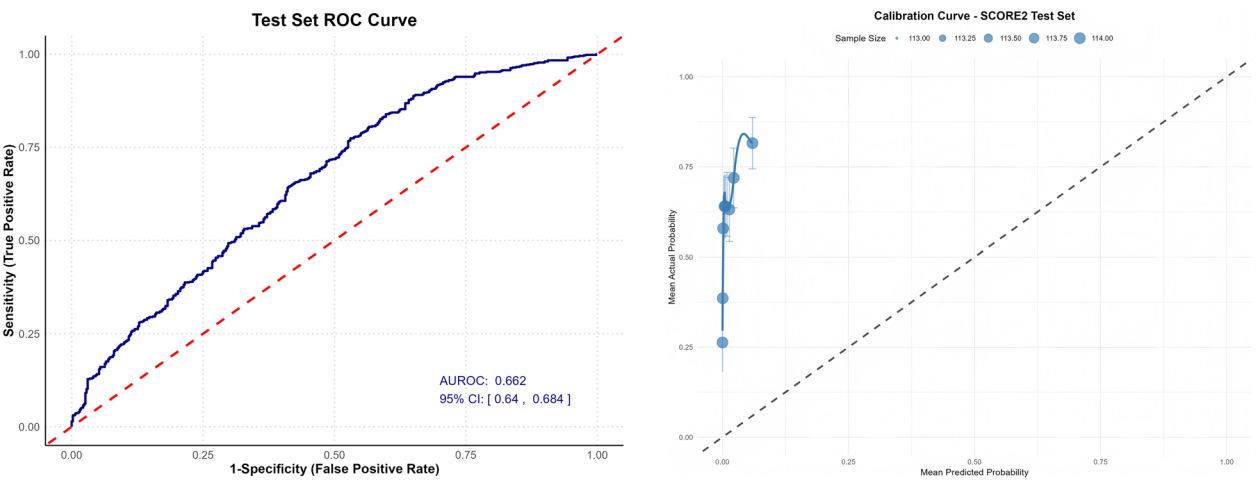


Fig.15 ROC and calibration curves of the SCORE2 Asia Pacific model in the male test set

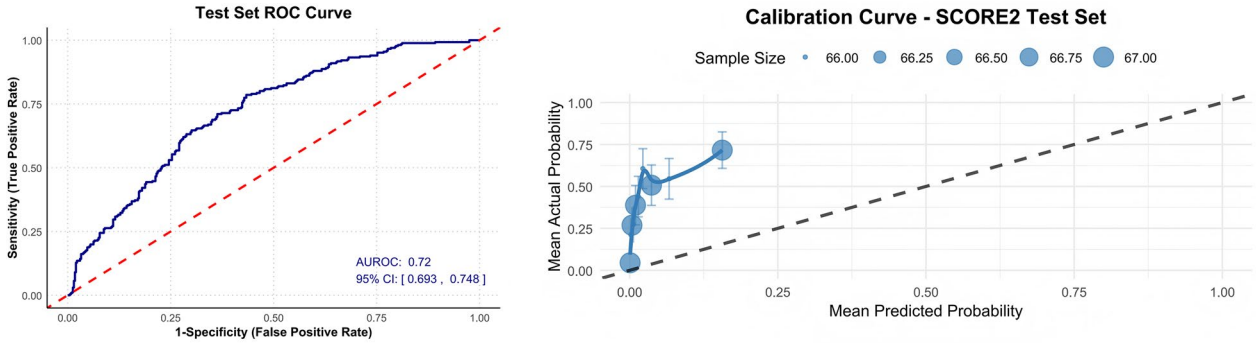


Fig.16 ROC and calibration curves of the SCORE2 Asia Pacific model in the female test set

validation, showing potential for improving CAD risk stratification and supporting clinical management decisions in China's Xinjiang region. Specifically, this study offers several key advantages that effectively address limitations observed in previous research in this area: first, the model is built upon a more representative multi-ethnic cohort and advanced methodology. Unlike previous studies in Xinjiang that were often constrained by small sample sizes, single-ethnic cohorts, or traditional statistical models [24, 37], this study utilized a larger cohort of over 12,000 individuals covering the four major ethnic groups (Han, Uyghur, Kazakh, and Hui). Furthermore, the machine learning algorithm we employed can capture complex nonlinear relationships, thereby enhancing the model's generalizability and robustness. Second, this research is based on high-quality, standardized clinical endpoints and incorporates novel biomarkers. Using the "gold standard"—coronary angiography or CTA—for all enrolled patients, this study ensured diagnostic reliability. Moreover, while previous local models often lacked key emerging risk factors, our model systematically includes the independent risk factor Lp(a) and, for the first time in such research, incorporates ethnicity as a key predictor. This helps compensate for heterogeneity arising from genetic and life style differences. Third, the model demonstrates excellent clinical practicality and convenience. The features included in the model, such as age, sex, ethnicity, basic vital signs, and routine blood biochemical indicators, are all easily obtainable in clinical practice. This non-invasive or minimally invasive characteristic enables the tool to be efficiently used for outpatient screening, aiding in the identification of high-risk individuals for targeted preventive interventions, thereby optimizing the allocation of healthcare resources. Finally, comprehensive validation and comparisons enhance the credibility of our model's results. We not only demonstrated the model's robustness through internal validation but also conducted a calibration comparison with the recalibrated SCORE2 Asia Pacific model. Additionally, we provided complete performance characteristics across all potential thresholds, offering a solid evidence base for the model's future clinical application and threshold selection.

This study has several limitations. First, to ensure data quality during the initial model exploration phase, we excluded patients with severe comorbidities, mental illnesses, or those who were pregnant. Although this criterion helped reduce confounding effects from complex conditions during the initial model development, it inevitably introduced selection bias and may limit the model's applicability in broader real-world populations with comorbidities. Future studies are needed to validate the model's generalizability in more heterogeneous cohorts that include these patient groups. Second, regarding the temporality of predictor variables, all clinical variables used for modeling were collected at initial patient admission prior to CAD diagnosis, ensuring their predictive nature. However, these data are inherently cross-sectional, obtained near the time of diagnosis, and do not capture long-term dynamic changes. Consequently, the model is more suitable for short-term risk assessment, and its long-term predictive performance and lead time require further validation in truly prospective studies incorporating multi-year follow-up data. Third, although our cohort included the four major ethnic groups in Xinjiang with proportions highly consistent with the regional demographic distribution, the total sample size remains relatively limited. While the model demonstrated encouraging discriminatory performance in Kazakh and Hui subgroups, estimates for these minority populations remain imprecise at the current sample size and require further validation in larger multi-ethnic cohorts. Additionally, we would have greatly preferred to compare our model with established models like China-PAR; however, due to the retrospective nature of our study, we lacked the

key "waist circumference" variable required by the CHINA-PAR model, making it impossible to compute and evaluate this model on our dataset. We have explicitly acknowledged this limitation in the Discussion section of the paper. Finally, regarding model validation, although internal validation demonstrated good discrimination and better calibration than existing models like the Asia Pacific-recalibrated SCORE2, the lack of an independent external validation cohort remains a critical limitation. The model's generalizability to external populations, along with its calibration stability and optimal decision thresholds, all require final confirmation in future prospective studies.

Conclusion

The robust performance of our model in both test and validation sets demonstrates its potential to enhance CAD risk classification and inform patient management strategies in clinical practice. Furthermore, the non-invasive nature of the proposed method, requiring only basic information (age, sex, marital status, etc.), blood pressure measurement, and blood sampling, enables efficient and straightforward patient selection for further diagnostic testing. Our findings indicate that the ML model can provide personalized and reasonably accurate CAD risk prediction. The interpretable insights may help identify various potentially modifiable risk sources for individual patients while reducing risks associated with invasive procedures, suggesting promising clinical applicability. Implementing our CAD prediction model in clinical practice is straightforward, as healthcare providers can easily integrate it into their workflow by inputting standard laboratory test results. This approach could significantly contribute to improving primary prevention and management of cardiovascular diseases among patients in Xinjiang.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s40001-025-03763-1.

Supplementary Material 1
Supplementary Material 2
Supplementary Material 3
Supplementary Material 4
Supplementary Material 5
Supplementary Material 6
Supplementary Material 7
Supplementary Material 8

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Not applicable.

Author contributions
G.M. and F.A. contributed equally to this work. All authors significantly contributed to this work. G.M., F.A., S.J., Q.Q.S., Z.Y.L. performed the research and collected the data; G.M., F.L., R.L, M.W.W and Y.Q.Z. analyzed the data; G.M., F.A. and Z.Y.F. designed the research study; G.M., Q.Q.S. and F.L. wrote the paper; and G.M. and F.A. prepared all the tables. All authors reviewed the manuscript. In addition, all authors approved the final draft.

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Data availability
All data generated or analyzed during this study are included in this published article.

Declarations

Ethics approval and consent to participate
This study complied with the Declaration of Helsinki and was approved by the Ethics Committee of the First Affiliated Hospital of Xinjiang Medical University (2022TSYCLJ0030; 2024E02043), and all participants signed informed consent forms. The study subjects are aware of and agree to the information to be published in an online open access publication. All text, icons, and tables do not contain the name of the patient or information that would allow the patient's identity to be obtained.

Consent for publication
All authors agreed that the study be published in this journal, and the manuscript has not been published and is not being considered for publication elsewhere.

Competing interests
The authors declare no competing interests.

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